

Kaiyang Li

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Education Background

- Ph.D. student, 2025–Present, Computer Science and Engineering, University of Connecticut
- Master, 2016, Electrical Engineering, University of Electronic Science and Technology of China
- Bachelor, 2013, Communication Engineering, University of Electronic Science and Technology of China

Internship Experience

Research Intern, Optical Networking & Sensing

NEC Laboratories America

Summer 2026

Mentor: Shaobo Han

- Developed the **Mizar Family**, compact audio-language models for acoustic situational awareness, including a 3B reasoning-on-demand model and a 160M always-on model.
- **Mizar Reasoning (3B)**: proposed Audio Advantage and Pair Audio Contrastive Learning for audio-grounded distillation; achieved 74.7 on MMAU Test-Mini, state-of-the-art among 3B audio-language models and matching a 7B model, with only 10K training examples and approximately one hour of training.
- **Mizar Thinking (160M)**: achieved state-of-the-art performance among sub-200M audio-language models; improved 3-benchmark average accuracy from 37.5 to 45.7 (+8.2) over the previous best sub-200M baseline while using fewer parameters (160M vs. 167M) and less compute (229 vs. 281 GFLOPs).

Selected Publications

1. **Kaiyang Li**, Shaobo Han, Qing Su, Wei Li, Zhipeng Cai, Shihao Ji. *Uni-LoRA: One Vector is All You Need*. **NeurIPS 2025 (Spotlight, top 3.19%)**. [Paper] [Code] [Hugging Face PEFT]
 - Designed and proposed Uni-LoRA, a unified and highly parameter-efficient framework that reconstructs LoRA parameters from a single shared vector via an isometric projection; achieved 0.0061% base-model-size tuning while matching or outperforming strong LoRA methods across multiple tasks.
 - Officially integrated into Hugging Face PEFT, Hugging Face’s open-source library for parameter-efficient fine-tuning, as a supported tuning method with official documentation, examples, tests, and save/load support.
2. **Kaiyang Li**, Shaobo Han, Qing Su, Shihao Ji. *ReForge: Refining Merged Models with Anchor-Regularized Regression*. **Submitted to ICLR 2027**. [Paper]

- Proposed ReForge, a bilevel framework that refines merged models module by module, formulating refinement as Bayesian linear regression with an anchor-centered prior (closed-form MAP from unlabeled calibration activations) and tuning it with Bayesian optimization.
 - ReForge consistently improves existing model-merging methods as a plug-and-play refinement.
3. **Kaiyang Li**, Shaobo Han, Yue Tian, Shihao Ji. *Mizar: A 159M-Parameter Audio-Language Model for Audio Understanding*. **Submitted to ICASSP 2027**. [Paper] [Code] [Model]
 4. **Kaiyang Li**, Shaobo Han, Yue Tian, Shihao Ji. *Reward-Tilted On-Policy Distillation for Acoustic Grounding in Audio-Language Models*. **Submitted to ICASSP 2027**. [Paper] [Code] [Model]
 5. **Kaiyang Li**, Guoming Lu, Guangchun Luo, Zhipeng Cai. *Seed-free Graph De-anonymization with Adversarial Learning*. In **CIKM (acceptance rate: 21%)**, 2020.
 6. **Kaiyang Li**, Guangchun Luo, Yang Ye, Wei Li, Shihao Ji, Zhipeng Cai. *Adversarial Privacy-preserving Graph Embedding against Inference Attack*. **IEEE Internet of Things Journal (IoT-J, Impact Factor: 9.94)**, 2020.

Peer Review

Reviewer for NeurIPS, ICLR, ICASSP, CIKM, IEEE IoT-J, IEEE TCSS, IEEE TIFS, HCC, EAAI, JIIS, AI Review, and Cluster Computing.

Awards & Honors

- Marion and Frederick Buckman Engineering Fellowship, UConn School of Computing, 2026.
- NeurIPS 2025 Scholar Award.

Work Experience

Applied Scientist

Sep. 2021 – Jul. 2022

Tencent Music Entertainment Group

- Developed machine learning models for hit song prediction using large-scale user behavior and audio feature data, and analyzed interaction patterns to support recommendation quality and user engagement.